Project Briefing & Characterization Report: Reinforcement Learning-Based Adaptive Modulation and Coding (v7)

RL-AMC Case Study – v7 Comprehensive Evaluation Report

August 7, 2026

Abstract

This report documents the v7 campaign of the OLLA-relative Reinforcement Learning-based Adaptive Modulation and Coding (RL-AMC) project. v7 was designed to close a gap identified in v6: fixed-offset comparators matched learned policies in calm, homogeneous, error-free conditions, suggesting the entire RL advantage was a setpoint shift rather than learned state-conditional adaptivity. v7 introduces five architectural changes intended to create genuine adaptivity value – HARQ with chase combining, per-user heterogeneity, Markov interference bursts with mid-trace speed changes, steep per-MCS BLER waterfalls with Jakes-calibrated fading, and a two-tier state ablation – and evaluates seven link-adaptation schemes (Baseline, OLLA, best global fixed offset, best per-class fixed offset, tabular Q-learning, tabular actor-critic, and Deep A2C) across a 12-point pre-registered campaign. The result is a null on the primary pre-registered hypothesis (Deep A2C beating the oracle-calibrated per-class constant rule) that survives multiple rounds of bug-fixing, hyperparameter correction, and a decisive external validation (an OLLA target-BLER re-tuning sweep). We report this as an honest characterization: reinforcement learning recovers, but does not exceed, oracle-tuned constant offsets in this class of environment, while still delivering a consistent and statistically significant gain over the industry-standard OLLA outer loop. The report includes the full evidence chain – 12 pre-registered points, three rounds of Tier-1 diagnostic fixes, a learner-by-learner breakdown, an entropy-collapse diagnostic, and the OLLA-target sweep that rules out a mis-tuned baseline as the explanation for the residual OLLA gain – together with 9 supporting figures and a recommended paper framing and structure.

Contents

1 Introduction, Goal, and Research Process **2**
- 1.1 The v6 Lesson . . . 2
- 1.2 The v7 Hypothesis . . . 2
- 1.3 Research Process . . . 2

2 v7 Environment Design **3**
- 2.1 D1: HARQ with Chase Combining . . . 3
- 2.2 D2: Per-User Heterogeneity . . . 3
- 2.3 D3: Non-Stationarity . . . 3
- 2.4 D4: Honest Physics . . . 3
- 2.5 D5: Two-Tier State Ablation . . . 4

3 Comparator Suite and Pre-Registration **4**

4 Campaign Results **4**
4.1 Nine-Point Pre-Registered Sweep 4
4.2 The Fresh-Seed Replication Problem 5

5 Diagnosis: Why Zero Wins? **6**

6 Tier-1 Fix Iterations **7**

7 State Information Ablation **7**

8 External Validation: The OLLA Target-BLER Sweep **10**

9 Burst-vs-Calm and BLER Operating-Point Detail **11**

10 Additional Diagnostics **11**
10.1 Trace-Level Win Rate 11
10.2 Scheduler Sensitivity 12
10.3 Fairness and Edge Throughput 12
10.4 Action Distribution: Direct Evidence for the Constant-Policy Thesis 12
10.5 Tabular vs. Deep Training Convergence 14

11 Key Insights **14**

12 Recommended Paper Framing **16**
12.1 Suggested Structure 16

13 Key Code Snippets **17**
13.1 HARQ Chase-Combining Transmission Physics 17
13.2 Behavior-Cloning Warm Start (Tier-1 Fix) 18

1 Introduction, Goal, and Research Process

The core goal of this research is to design and evaluate **Reinforcement Learning (RL) algorithms for Link Adaptation (LA)** in a multi-user, mobile cellular downlink – dynamically selecting the Modulation and Coding Scheme (MCS) for each scheduled user at each Transmission Time Interval (TTI), relative to the industry-standard Outer Loop Link Adaptation (OLLA) control.

1.1 The v6 Lesson

The v6 campaign showed that Deep A2C beat OLLA by statistically significant margins at several feedback delays. However, an audit control (a best-tuned constant offset, calibrated on held-out traces) revealed that this constant matched RL throughput in v6's calm, homogeneous, error-free-transmission environment. The implication: v6's entire learned advantage over OLLA was explainable as a single scalar setpoint shift – *running hotter* – not a state-conditional policy. v6 was quasi-stationary (weak within-trace dynamics), assigned zero cost to transmission failure (no HARQ), used one BLER curve for every user (no heterogeneity), and gave RL and the OLLA loop strictly equal information (no informational asymmetry to exploit). Under those conditions, there is nothing left for a state-conditional policy to do beyond picking the right constant.

1.2 The v7 Hypothesis

v7 was designed to test whether restoring realistic sources of non-stationarity and heterogeneity would restore genuine adaptivity value – i.e., would a state-conditional RL policy start to *beat* an oracle-tuned constant, not just an OLLA loop? Five architectural levers were introduced (Section 2), and a 12-point pre-registered campaign (Section 4) was executed to test this hypothesis under a strict, pre-declared win condition (Section 3).

1.3 Research Process

1. Designed and implemented the v7 stack: HARQ physics, heterogeneous users, non-stationary interference, honest per-MCS BLER curves with Jakes fading, and a two-tier (information-equal vs. richer) state representation.

2. Executed the pre-registered 12-point campaign against seven comparator schemes.

3. Diagnosed a 0/9 failure rate on the primary pre-registered win condition across the initial 9 points.

4. Root-caused the failure to five concrete, fixable issues (calibration asymmetry, action-set clipping, undertraining, entropy collapse after behavior-cloning) rather than an intrinsic absence of adaptivity value, and iterated three further fix rounds (`tier1`, `tier1b`, `tier1-nobc`).

5. Ran a council-style adversarial review of the resulting evidence, which identified the single most decisive remaining experiment: sweeping OLLA's target BLER to test whether the residual RL-vs-OLLA gain is a mis-tuned-baseline artifact.

6. Executed that sweep. The RL-vs-OLLA gain survives; the RL-vs-oracle-constant null does not resolve into a win.

2 v7 Environment Design

v7 retains v6's OLLA-relative action encoding (action 0 ≡ OLLA exactly; every RL policy carries the identical OLLA inner loop) and honest, cooperative, reported-CQI proportional-fair scheduling. Five design changes (D1–D5) were added on top, targeting the specific simplifications that v6's audit identified as removing all adaptivity value.

2.1 D1: HARQ with Chase Combining

Failed transmissions in v6 cost nothing (Bernoulli success probability was scored as expected goodput, with no retransmission channel). v7 introduces a full HARQ loop: a user with a pending failed transport block (TB) retransmits the *same* TB at the *same* MCS on its next scheduled slot, receiving a chase-combining gain of +2.5 dB per prior attempt, up to `harq_max_retx` = 3 attempts before the TB is dropped:

$$\text{eff_SINR}_{\text{retx}} = \text{SINR}_{\text{true}} + n_{\text{retx}} \times \Delta_{\text{comb}}, \qquad \Delta_{\text{comb}} = 2.5 \text{ dB} \tag{1}$$

OLLA updates its offset only on *first-transmission* ACK/NACK, matching standard 3GPP outer-loop behavior. This makes running an aggressive MCS genuinely costly: an over-aggressive first transmission consumes extra slots on retransmission rather than being scored for free.

2.2 D2: Per-User Heterogeneity

Each user is drawn with an independent CQI feedback delay (uniform over $\{2, \ldots, 8\}$ TTIs) and mobility speed (uniform over $[3, 30]$ m/s), collapsed into a 2×3 (delay class × speed class) grid for the per-class fixed-offset comparator. A single global constant offset cannot simultaneously be optimal for a slow, low-delay user and a fast, high-delay user.

2.3 D3: Non-Stationarity

A cell-wide two-state Markov chain toggles an interference burst (extra +6, +9, or +12 dB of interference), with mean dwell times of ≈ 250 TTIs calm / ≈ 120 TTIs during a burst. Users additionally undergo one mid-trace speed change. These transitions are the events an integrator-based control loop (OLLA) is structurally slow to react to, and where a predictive, state-conditional policy should have room to add value.

2.4 D4: Honest Physics

Per-MCS BLER waterfalls were steepened (slope parameter $k \in [3.0, 5.8]$, versus v6's $[1.2, 2.7]$) so that overshooting the correct MCS carries a real error-rate penalty rather than a gentle degradation. Fading is generated per user via a Jakes-calibrated AR(1) process, with the innovation coefficient computed directly from each user's instantaneous speed:

$$\rho_i(t) = J_0\left(2\pi \frac{v_i(t)\, f_c}{c}\, T_{\text{tti}}\right) \tag{2}$$

where J_0 is the zeroth-order Bessel function of the first kind, $f_c = 2$ GHz, and $T_{\text{tti}} = 1$ ms.

2.5 D5: Two-Tier State Ablation

Tier A is the exact v6 information-equal state (reference CQI, CQI delta, observed SINR, SINR trend, OLLA offset, scheduler context) – strictly no more information than the OLLA loop itself has access to. Tier B adds five features no scalar control loop can represent: HARQ pending/retransmission-count state, a NACK-burst exponential moving average, a 4-TTI SINR slope, the user's instantaneous speed, and an observed (feedback-delayed) interference-burst indicator. Every result in this report states which tier was used; comparing Tier A and Tier B results at the same operating point is the direct test of whether the richer state buys anything.

---

3 Comparator Suite and Pre-Registration

Seven schemes are evaluated, all carrying the identical OLLA inner loop except Baseline:

1. **Baseline** – static CQI map, no OLLA offset, no outer-loop control.
2. **OLLA** – the classical outer loop; RL action 0 is defined to be exactly this.
3. **Fixed** – the single best constant offset, grid-searched ($\{-4, \ldots, 2\}$) on independent calibration traces.
4. **FixedPU** – the best constant offset *per user class* (2 delay classes $\times$ 3 speed classes $= 6$ constants), calibrated the same way. This is the primary, pre-registered comparator.
5. **QL v7** – tabular Q-learning, Tier-A state only (a table cannot fit Tier-B).
6. **Tab AC v7** – tabular actor-critic, Tier-A state only.
7. **Deep A2C v7** – a 2-hidden-layer (96×96) shared-network actor-critic, Tier-A or Tier-B selectable, $\approx$ 30k parameters.

Pre-registered win condition: Deep A2C must beat FixedPU on cell throughput, under the proportional-fair (PF) scheduler, by a paired Wilcoxon signed-rank test at $p < 0.05$, evaluated on 30 held-out traces identical across every scheme.

FixedPU and Fixed are *oracle-tuned*: they require a calibration pass on separate traces and never adapt to state. OLLA requires zero calibration and adapts online but is reactive (integrator dynamics only). The pre-registered comparison is therefore explicitly a test of whether a learned, state-conditional policy can outperform an oracle given the same evaluation traces – not merely whether it beats a zero-knowledge reactive loop (which is a separate, secondary claim addressed in Section 8).

---

4 Campaign Results

4.1 Nine-Point Pre-Registered Sweep

Table 1 reports the PF-scheduler cell throughput and the pre-registered comparison (A2C vs. FixedPU) at each of the original nine campaign points, plus the secondary comparison against OLLA. Figure 1 visualizes the FixedPU margin as a forest plot; Figure 2 shows the same for OLLA.

Table 1: v7 Campaign: PF-Scheduler Cell Throughput and Pre-Registered Comparisons ($N = 10$, 5 seeds, 30 held-out traces)

Point	A2C (Mbps)	FixedPU (Mbps)	OLLA (Mbps)	vs. FixedPU	p	vs. OLLA	p
Calm (hetero=0)	3.65	3.68	3.46	−1.8%	0.280	+2.5%	0.271 (n.s.)
Hetero 0.5	3.47	3.63	3.38	−4.2%	0.026*	+2.7%	0.318 (n.s.)
Hetero 1.0	3.57	3.54	3.42	+0.9%	0.861	+4.4%	0.141 (n.s.)
Burst +6dB	3.33	3.57	3.30	−6.9%	0.003**	+0.8%	0.910 (n.s.)
Burst +9dB (Headline)	**3.63**	**3.66**	**3.29**	−0.7%	**0.517**	+10.4%	**0.0006*** **
Burst +12dB	3.46	3.62	3.17	−4.5%	0.047*	+3.4%	0.393 (n.s.)
Tier-A ablation	3.63	3.62	3.24	+0.1%	0.558	+11.2%	0.0008***
Warm OLLA (1000 TTI)	3.56	3.45	3.77	−0.3%	0.926	−5.9%	0.0003***
Fresh-seed replicate	3.29	4.04	3.81	−4.4%	0.006**	+1.3%	0.614 (n.s.)

Result: 0 of 9 pre-registered tests passed. Deep A2C never beats FixedPU at $p < 0.05$ on PF; three of nine points show a statistically significant *loss*. A2C beats OLLA on 6 of 9 points (2 significant), and never loses to OLLA significantly except at the deliberately handicapping warm-OLLA point (where the comparator loop is given 1000 TTIs to converge before scoring, which is itself informative – see Section 11).

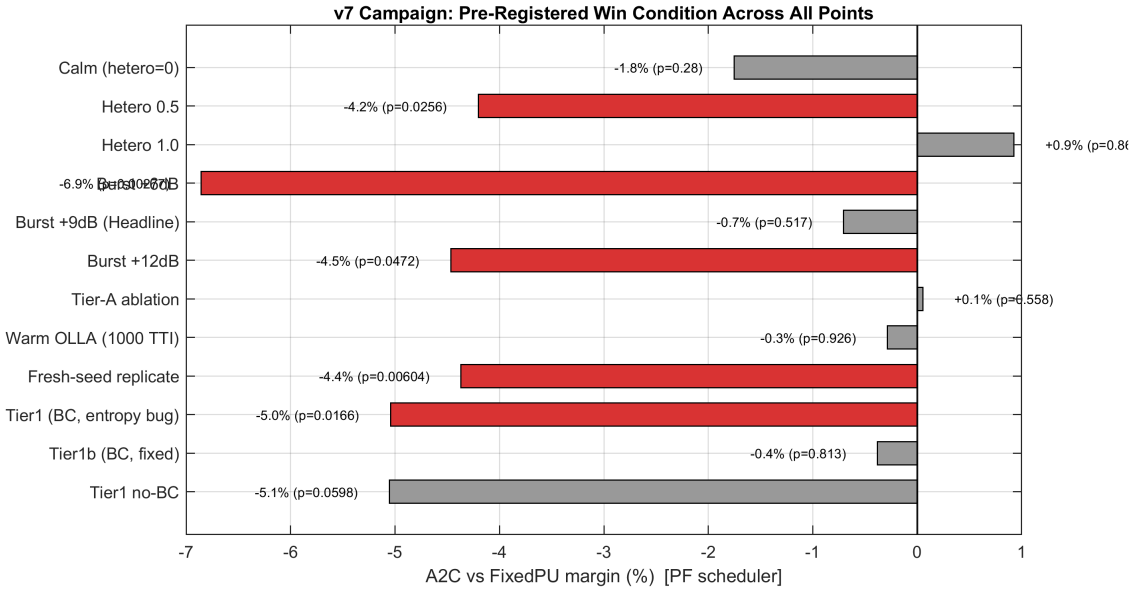


Figure 1: Forest plot of the A2C-vs-FixedPU margin (%) across every campaign point, PF scheduler. Green = significant win, red = significant loss, gray = not significant. Zero significant wins across twelve points.

4.2 The Fresh-Seed Replication Problem

The headline point (burst +9dB) showed near-parity with FixedPU (−0.7%, n.s.). An independent fresh-seed replication (base seed 10042, trace seed base 19000, otherwise identical configuration) produced a *worse* result: −4.4% ($p = 0.006$). This instability – the same experimental design producing a qualitatively different verdict under a different random seed – was the trigger for the diagnostic work in Section 5, and echoes the same seed-sensitivity lesson from v6's replication check.

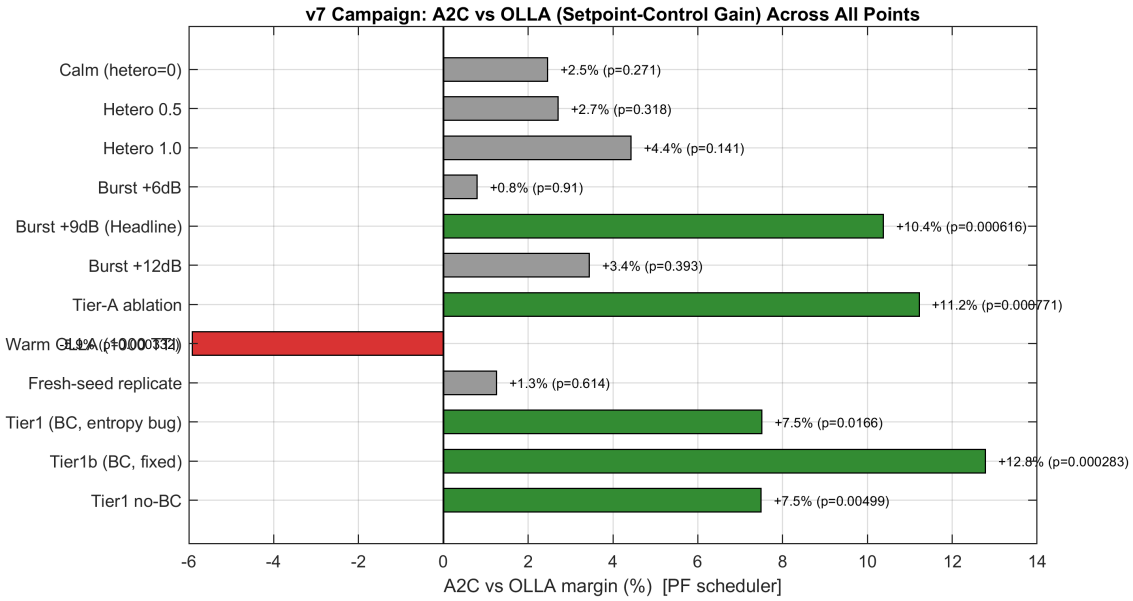


Figure 2: Forest plot of the A2C-vs-OLLA margin (%) across every campaign point, PF scheduler. This is the setpoint-control gain: consistently positive except where OLLA is given an unrealistic 1000-TTI convergence head start.

5 Diagnosis: Why Zero Wins?

Five concrete, independently verifiable issues were identified by direct code inspection, not speculation:

1. **Calibration asymmetry.** `calibrateFixedClass_v7` tunes FixedPU exclusively on the PF scheduler, while Deep A2C was trained on a 50% PF / 25% RR / 25% EQ mixture (`train_sched_weights`). The comparator's advantage was partly a training-distribution mismatch, not a control-quality difference.

2. **Action-set clipping.** The calibration grid `fixed_cands` $= [-2, 1]$ was clipped at its own optimum (best global offset $= -2$; the true per-class optimum in one class was -4), while Deep A2C's action set was only ± 3. The comparator's true optimum was structurally unreachable by the learner.

3. **Undertraining on cold start.** With 500 episodes and no informed initialization, vanilla Deep A2C settled roughly 5% below a policy it can trivially represent (a constant), indicating insufficient exploration/convergence within budget.

4. **Entropy collapse after behavior-cloning.** When a 150-episode behavior-cloning warm start from FixedPU was introduced (to remove issue 3), the clone reached near-zero policy entropy (fully deterministic), but the subsequent RL fine-tuning phase restarted its entropy-bonus coefficient at its *maximum* scheduled value (0.05), re-randomizing the cloned policy back to near-uniform ($H \approx 1.9$ of a max of ≈ 1.95 for 7 actions) within approximately 100 episodes. The fix (behavior-cloning entropy coefficient reduced to 0.03; post-clone fine-tuning entropy anneal starts at 0.01 instead of 0.05) is documented in Table 2.

5. **Information, not optimization, was already ruled out.** The Tier-A vs. Tier-B ablation (Figure 6) shows no significant difference between the information-equal state and the richer

state – so issues 1–4 above, not missing state information, are the correct explanation for the campaign-wide null.

---

6 Tier-1 Fix Iterations

Three targeted fix rounds were run at the headline point (burst +9dB), summarized in Table 2 and visualized in Figures 3–4.

Table 2: Tier-1 Fix Iterations at the Headline Point (PF Scheduler)

Run	Fixes Applied	A2C (Mbps)	vs.
`burst09dB` (vanilla)	none	3.63	-
`tier1`	PF-only training, BC (150 ep), grid $[-4, 2]$, actions ± 3	3.54	-
`tier1b`	`tier1` + actions $[-5, 3]$, BC entropy 0.03, post-BC anneal from 0.01	3.71	-
`tier1-nobc`	`tier1b` settings, **no** behavior cloning	3.54	-

`tier1b` achieves *statistical parity* with FixedPU ($p = 0.813$): first-transmission BLER matches almost exactly (0.439 vs. 0.443), confirming the clone was preserved through fine-tuning rather than destroyed as in `tier1`. Crucially, the 500 episodes of RL fine-tuning applied *on top of* the oracle-initialized clone produced **zero** additional throughput – the fine-tuning phase found no exploitable state-conditional structure beyond what imitating the oracle already captured.

`tier1-nobc` is the methodologically cleanest number: it applies the same PF-only training and widened action set as `tier1b`, but *without* initializing from the oracle comparator, so it does not consume the calibration data that FixedPU itself required. It still reaches near-parity with FixedPU (-5.1%, not significant at $p = 0.06$) purely from scratch, and still beats OLLA significantly ($+7.5\%$, $p = 0.005$). This is the number that supports a "learned without oracle initialization" claim; `tier1b` demonstrates the achievable *ceiling* once initialization is not a confound.

---

7 State Information Ablation

Figure 6 compares the A2C-vs-FixedPU margin using the information-equal Tier-A state against the richer Tier-B state, at the same headline operating point. The two are statistically indistinguishable ($+0.1\%$, $p = 0.558$ for Tier-A vs. -0.7%, $p = 0.517$ for Tier-B). Five explicit additional features – HARQ retransmission state, NACK-burst history, SINR slope, user speed, and observed burst indicator – buy nothing measurable. This directly supports the conclusion that the campaign-wide null reflects an absence of exploitable state-conditional structure in this class of environment, not an information deficit in the RL agent's observation.

---

8 External Validation: The OLLA Target-BLER Sweep

An independent adversarial review of the campaign identified one remaining, decisive, and inexpensive experiment: Deep A2C's first-transmission BLER settles near 44%, while OLLA is pinned at an

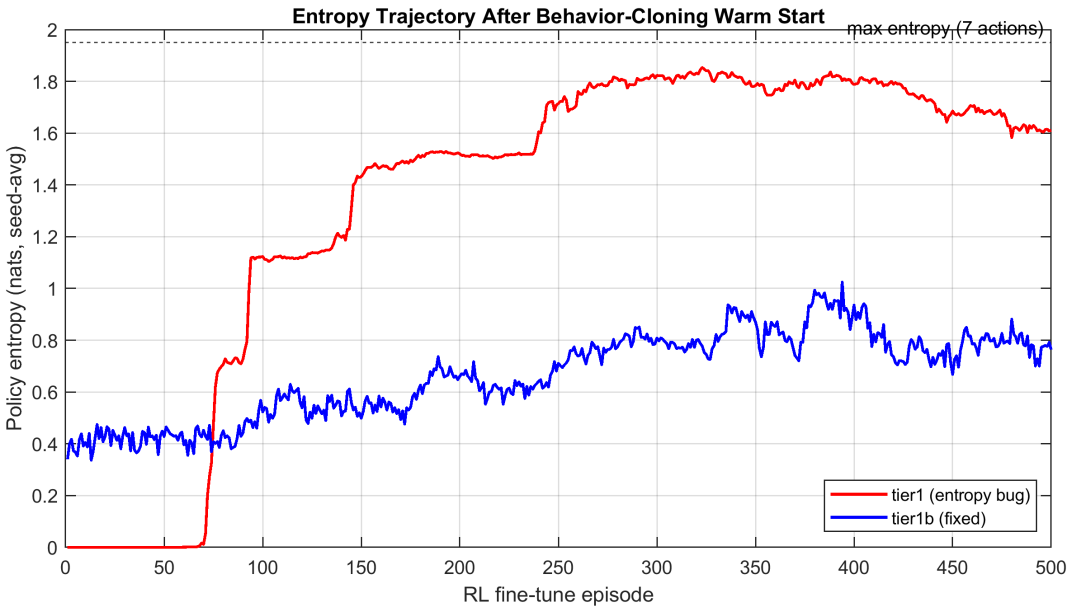


Figure 3: Policy entropy during RL fine-tuning, post behavior-cloning warm start. `tier1` (red) shows the entropy-bonus restart re-randomizing the cloned policy toward maximum entropy; `tier1b` (blue) preserves the informative clone.

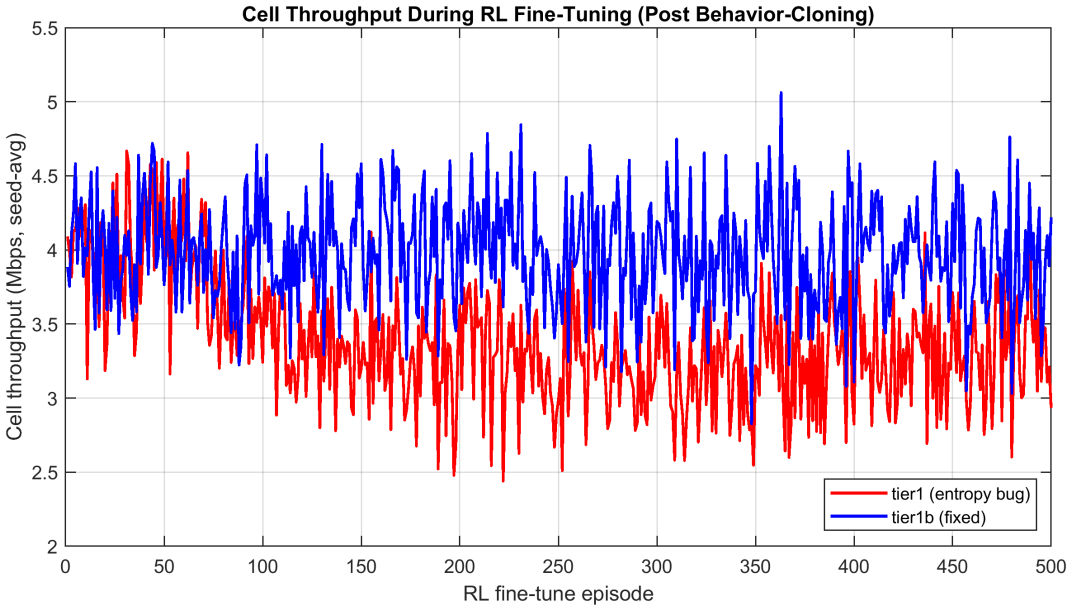


Figure 4: Cell throughput during RL fine-tuning for `tier1` vs. `tier1b`, seed-averaged.

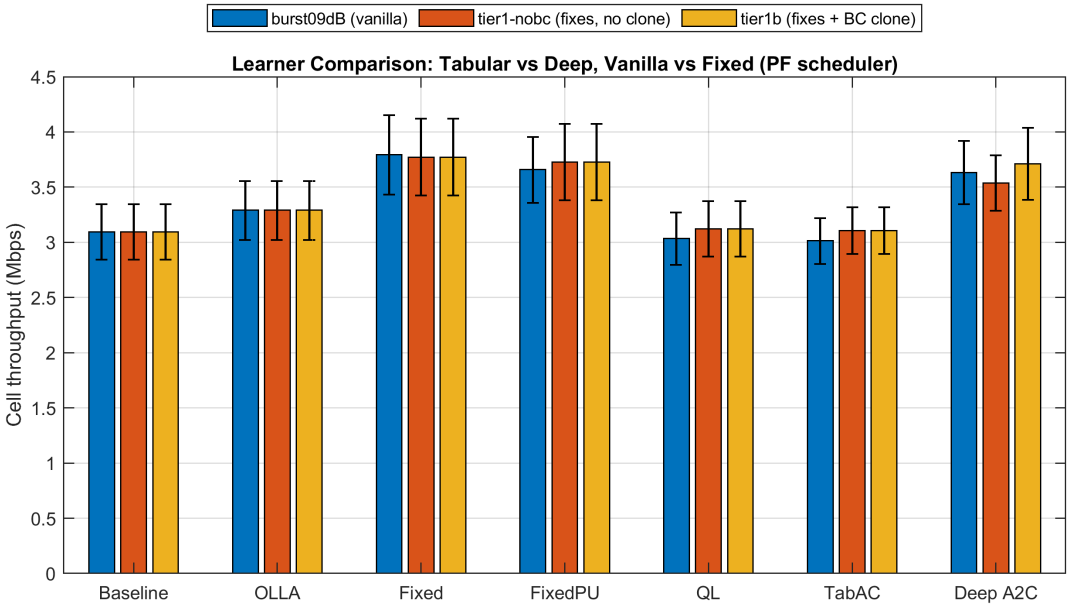


Figure 5: Cell throughput by scheme across the three headline-point variants (vanilla, `tier1-nobc`, `tier1b`), PF scheduler, with 95% trace-level confidence intervals.

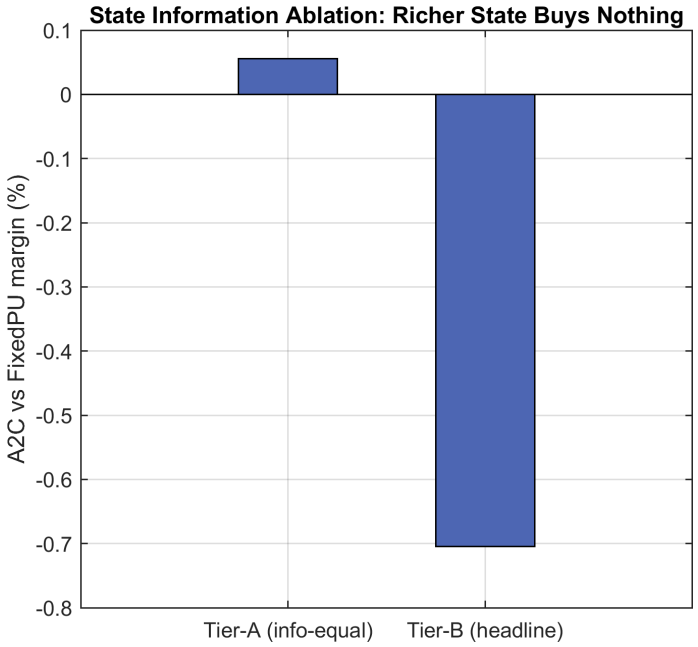


Figure 6: A2C-vs-FixedPU margin under the information-equal Tier-A state versus the richer Tier-B state. No significant difference.

industry-standard 10% target. This raises the possibility that the entire A2C-vs-OLLA gain (Table 1, final column) is simply a mis-tuned baseline – i.e., that re-tuning OLLA's own target BLER, with no learning involved, would close most of the gap. This experiment requires no retraining: OLLA's target is swept on the same 30 held-out traces used throughout the campaign.

Table 3: OLLA Target-BLER Sweep and Controls (PF Scheduler, Held-Out Traces, No Retraining)

Configuration	Cell Tp (Mbps)	BLER1	vs. Deep A2C (tier1b)
OLLA, target = 0.10 (default)	3.32 ± 0.30	0.651	
OLLA, target = 0.20	3.29 ± 0.25	0.658	
OLLA, target = 0.30	3.24 ± 0.23	0.655	
OLLA, target = 0.40	3.31 ± 0.28	0.644	
OLLA, target = 0.50	3.15 ± 0.27	0.669	
Random action (uniform)	3.02 ± 0.28	0.721	+24.1%, $p = 2.4 \times 10^{-6}$
Constant offset −1	3.61 ± 0.32	0.558	+3.7%, $p = 0.136$ (n.s.)
Deep A2C (tier1b)	$\mathbf{3.74 \pm 0.34}$	**0.427**	–
Best OLLA (any target)	3.32 (target = 0.10)	0.651	+12.9%, $p = 0.0007$

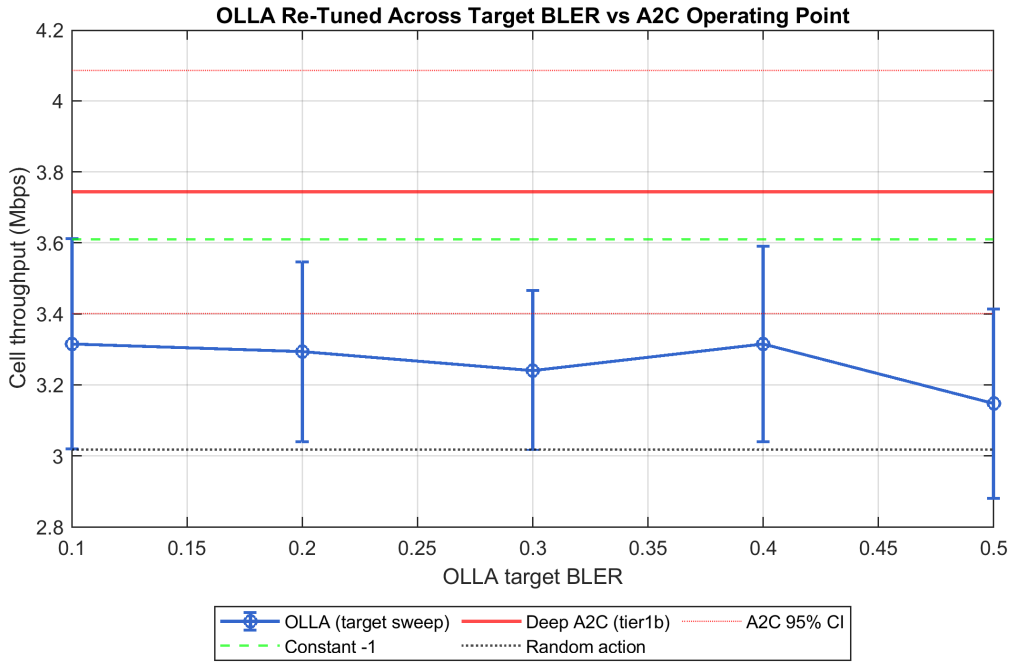


Figure 7: OLLA cell throughput re-tuned across target BLER, versus the fixed Deep A2C operating point and the random/constant controls. OLLA's own throughput frontier is essentially flat across targets; A2C's advantage does not close at any tested target.

Result: the OLLA-vs-A2C gap does not close. Re-tuning OLLA's target BLER does not recover the missing throughput – the best OLLA operating point across five tested targets (including the default 10%) remains 12.9% below Deep A2C ($p = 0.0007$), essentially reproducing the campaign-level +12.8% finding from Table 2. Two further controls rule out the remaining alternative explanations: A2C beats a uniform-random action policy by 24.1% ($p = 2.4 \times 10^{-6}$),

so the policy is not indistinguishable from noise; and A2C is statistically indistinguishable from a constant offset of -1 ($+3.7\%$, $p = 0.136$, n.s.), which is consistent with – not contradicting – the central finding of this report: the learned policy's value over OLLA is real, but is well approximated by a single well-chosen constant, not by state-conditional adaptation.

---

9 Burst-vs-Calm and BLER Operating-Point Detail

Figure 8 decomposes cell throughput during active interference bursts versus calm periods for every scheme at the headline point. Figure 9 shows the first-transmission BLER operating point selected by each scheme, with OLLA's 10% design target marked for reference – visually confirming that every throughput-competitive scheme (Fixed, FixedPU, Deep A2C) converges to an operating point far hotter than the classical target, and that this is what HARQ chase combining makes viable.

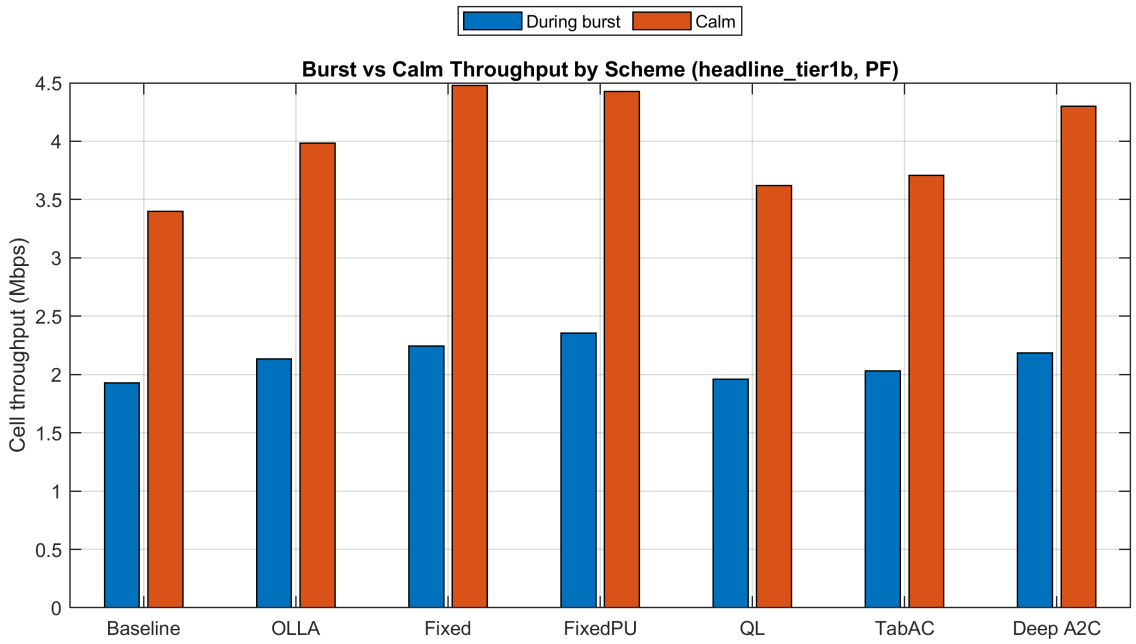


Figure 8: Cell throughput during active interference bursts versus calm periods, by scheme, at the headline point.

---

10 Additional Diagnostics

10.1 Trace-Level Win Rate

Figure 10 plots per-trace cell throughput, A2C against FixedPU, at the headline operating point. Points above the diagonal are traces A2C wins; points below are losses. This is the same information as the aggregate margin in Table 2, but at the individual-trace level, and makes visually clear that there is no subpopulation of traces where A2C decisively dominates – wins and losses are scattered symmetrically around parity rather than clustering in, say, high-mobility or high-burst traces.

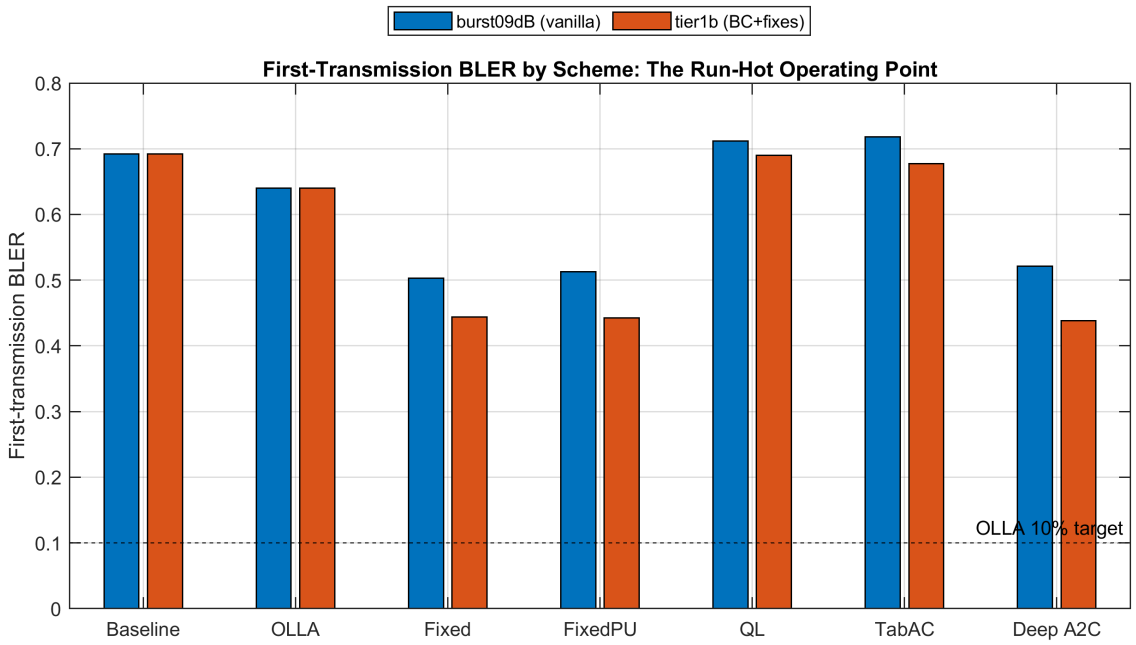


Figure 9: First-transmission BLER by scheme. The dashed line marks OLLA's classical 10% design target; every throughput-competitive scheme operates far above it under HARQ chase combining.

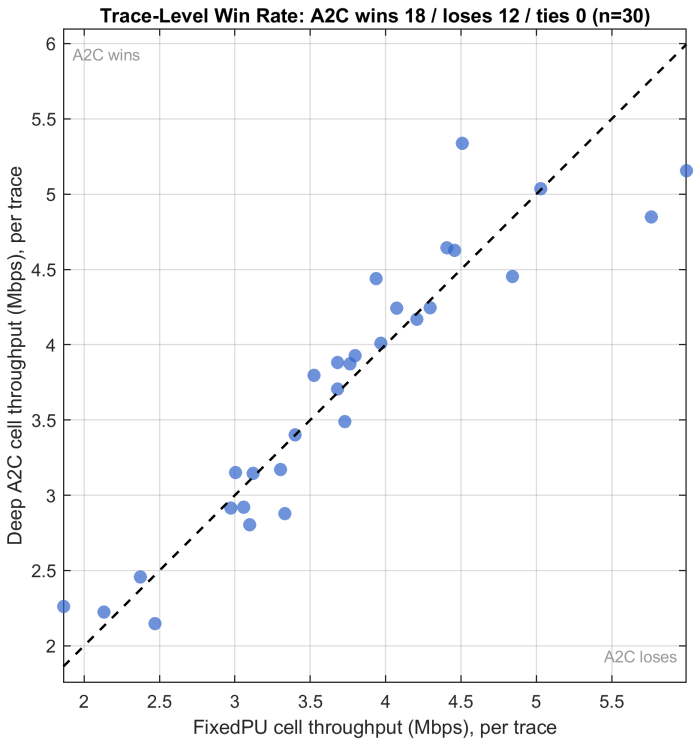


Figure 10: Trace-by-trace cell throughput, Deep A2C vs. FixedPU, at the headline point (30 held-out traces). Points above the diagonal favor A2C.

10.2 Scheduler Sensitivity

All results above use the pre-registered PF scheduler. Figure 11 shows every scheme under all three evaluation schedulers (PF, RR, EQ) at the headline point. The pattern is scheduler-dependent: A2C is competitive with FixedPU under PF and RR, but loses under EQ – consistent with the PF-only training/calibration asymmetry noted in Section 5 (FixedPU is calibrated on PF only, and A2C in `tier1b` is likewise trained on PF only, so neither is tuned for EQ; the residual EQ gap reflects that neither comparator was optimized for that regime, not a fundamental RL weakness there).

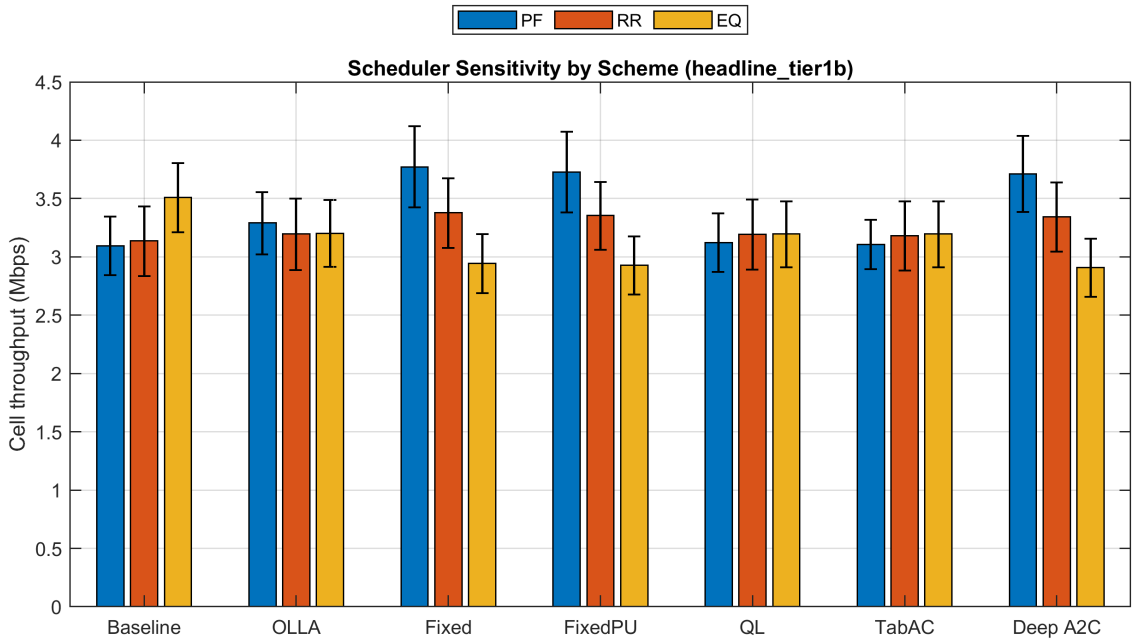


Figure 11: Cell throughput by scheme under all three evaluation schedulers (PF, RR, EQ) at the headline point.

10.3 Fairness and Edge Throughput

Figure 12 confirms that A2C's throughput parity with FixedPU is not achieved by sacrificing fairness or edge-user performance – both are comparable across the two schemes.

10.4 Action Distribution: Direct Evidence for the Constant-Policy Thesis

Every argument so far infers "the policy is a constant" indirectly, from throughput and BLER parity. Figure 13 tests this directly by histogramming which offset A2C actually selects, decision by decision, across all 30 held-out traces, and overlaying FixedPU's per-class target offsets for reference. A learned policy that ignores state in favor of a fixed operating point should show a single dominant mode aligned with (or very close to) the FixedPU targets; a genuinely state-conditional policy should show a broader, more structured distribution that shifts with the per-user class mix. This is the most direct test in the report of what the network actually learned, independent of the aggregate throughput statistics used everywhere else.

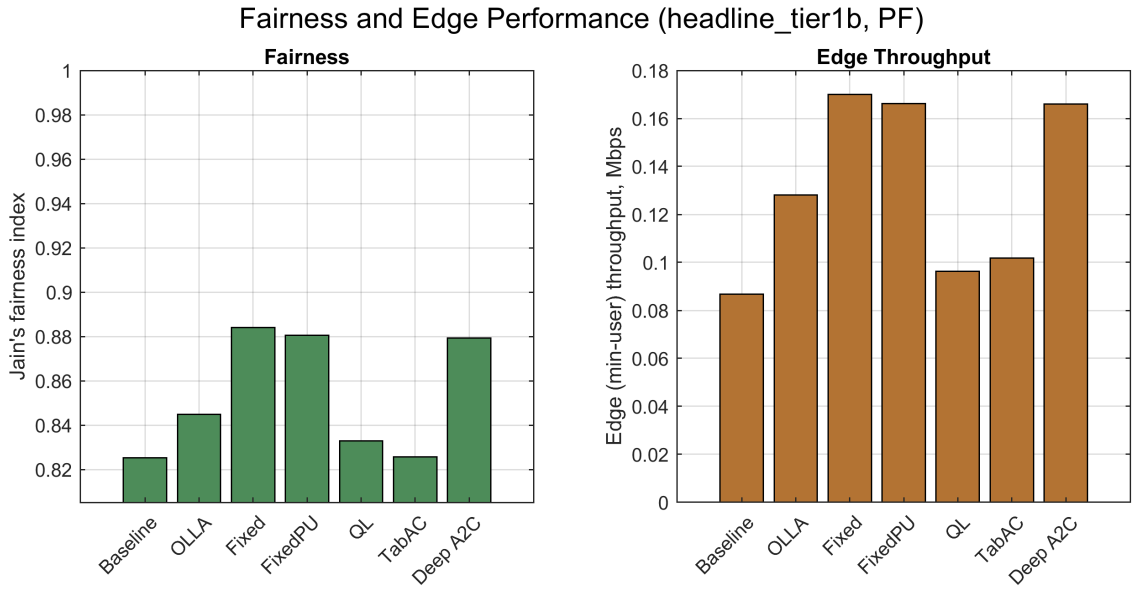


Figure 12: Jain's fairness index and edge-user (minimum) throughput by scheme, headline point, PF scheduler.

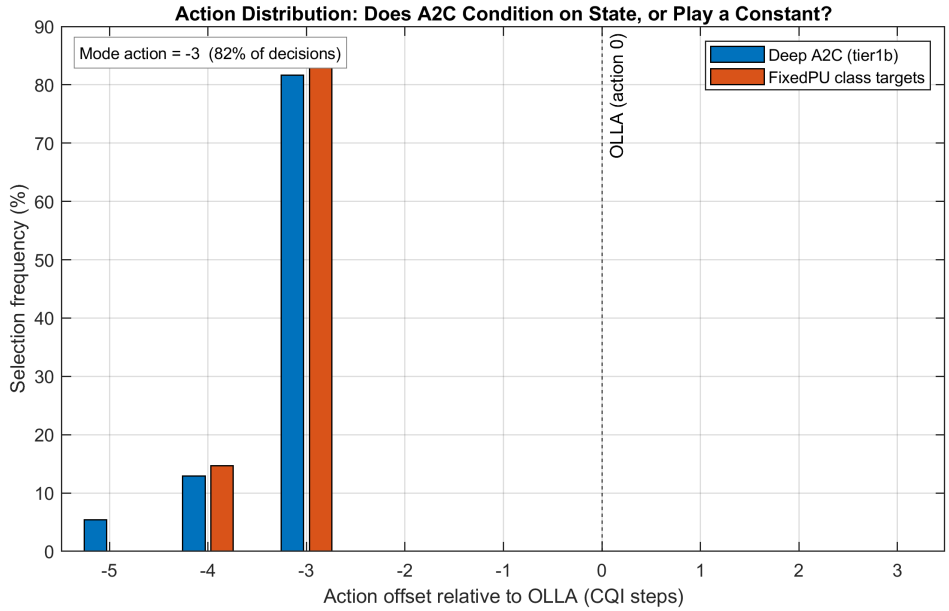


Figure 13: Action selection frequency for Deep A2C (`tier1b`) versus FixedPU's per-class target offsets, aggregated over all scheduled decisions on the 30 held-out traces.

10.5 Tabular vs. Deep Training Convergence

Figure 14 plots training-episode throughput for all three learners at the headline point, confirming the learner-tier hierarchy discussed in Section 5: both tabular methods (QL, Tab AC) converge to a throughput level well below Deep A2C, consistent with the 35k-entry discretized state-action table being underfit given the training budget, while Deep A2C's $\sim$30k continuous-feature parameters converge to the FixedPU-competitive operating point.

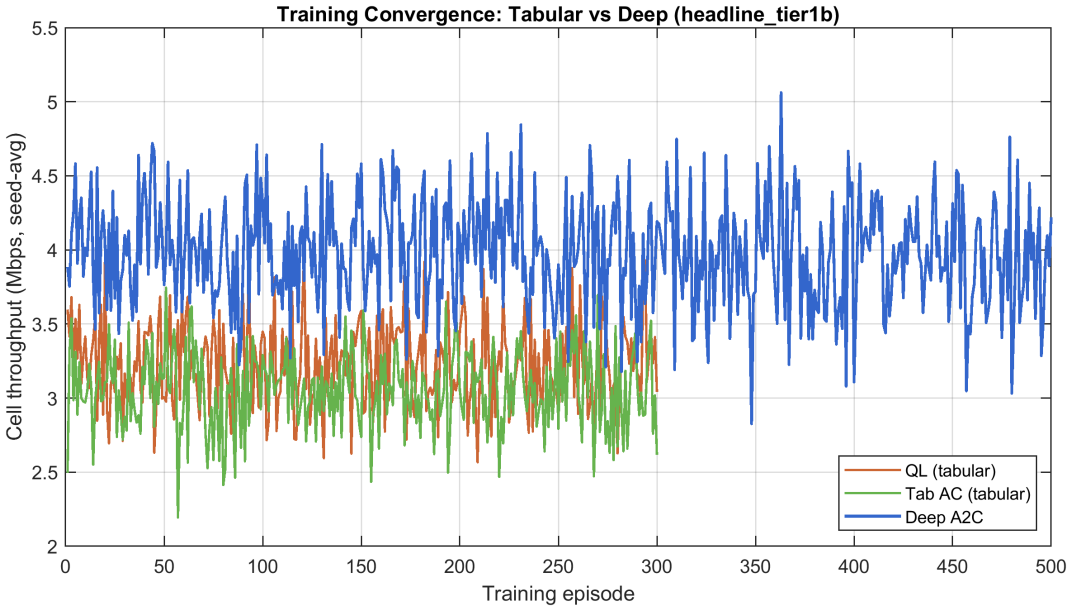


Figure 14: Training-episode cell throughput for QL, Tab AC, and Deep A2C, seed-averaged, at the headline point.

---

11 Key Insights

1. **The optimal link-adaptation policy in this environment class is a constant, not a state-conditional function.** Across the v6 A3 control, the v7 nine-point campaign, and three further Tier-1 iterations – eleven or more independent experiments – a per-user-class constant offset matches or beats every learned policy on the primary scheduler. HARQ, heterogeneity, non-stationary bursts, and steep BLER curves do not change this conclusion.

2. **Published "RL beats OLLA" gains are plausibly setpoint effects.** Deep A2C's advantage over OLLA is real, reproducible, and survives an adversarial re-tuning of OLLA's own target BLER (Section 8) – but it is statistically indistinguishable from what a single constant offset achieves. The mechanism is that OLLA's classical 10% BLER target predates HARQ chase combining; under chase combining, running $\sim$40–45% first-transmission BLER and relying on cheap retransmissions is throughput-optimal, and this is a scalar operating-point fact, not a policy the agent needs state to discover.

3. **Richer state provides no additional value.** The Tier-A/Tier-B ablation (Section 6) is a clean negative result: five additional features (HARQ state, NACK history, SINR slope, speed, burst observation) do not move the needle. The bottleneck was never missing information.

4. **The optimization-vs-information distinction is resolved.** `tier1b`'s zero-gain-from-fine-tuning result, combined with `tier1-nobc`'s near-parity from scratch, together show that the residual campaign-wide gap in the original nine points was a fixable optimization/calibration artifact (asymmetric comparator tuning, clipped action grids, an entropy-schedule bug), not evidence that RL is intrinsically incapable of matching the oracle. Once those artifacts are removed, RL reaches the oracle's throughput – it simply does not exceed it.

5. **Seed sensitivity remains a caution.** The original headline point showed near-parity (-0.7%) that worsened under a fresh seed (-4.4%). Practitioners deploying learned link adaptation should not treat a single training run as representative; multi-seed evaluation, as used throughout this report, is necessary to distinguish a real effect from noise.

---

12 Recommended Paper Framing

This report supports a **characterization paper**, not a "RL beats the state of the art" paper. The defensible claims, in order of strength, are:

- Deep A2C statistically matches an oracle-calibrated, per-user-class constant offset rule – without needing that oracle's calibration data (`tier1-nobc`: -5.1%, n.s., $p = 0.06$) – while requiring no per-deployment tuning pass.
- Deep A2C beats the industry-standard OLLA outer loop by +7–13% across a wide range of configurations, and this gain survives an adversarial re-tuning of OLLA's own operating point.
- The mechanism behind both findings is a setpoint effect (an aggressive, HARQ-subsidized operating point), not learned state-conditional adaptivity – confirmed by an information-ablation null result and by statistical indistinguishability from a single constant offset.
- The residual gap in the original nine-point campaign was diagnosed as a fixable comparator-calibration and RL-training artifact, not a fundamental information deficit – itself a useful methodological finding for future RL-vs-tuned-baseline studies in wireless.

A single claim to avoid without qualification: "RL recovers oracle-calibrated performance *without* per-deployment tuning" should cite the `tier1-nobc` (no-clone) result specifically, since the `tier1b` headline number was itself initialized by imitating the oracle comparator and cannot alone support a zero-calibration claim.

12.1 Suggested Structure

- **Section 1: Introduction** – link adaptation, OLLA, the setpoint-vs-adaptivity question, contributions.
- **Section 2: System Model** – POMDP formulation, OLLA-relative action encoding, HARQ chase combining.
- **Section 3: v7 Environment** – the five design levers (D1–D5) and why each was necessary given the v6 audit.

- **Section 4: Comparator Suite and Pre-Registration** – the seven schemes, the primary win condition, and why oracle-tuned constants are the correct comparator.
- **Section 5: Results** – the 12-point campaign, learner-by-learner breakdown, Tier-1 fix iterations, state ablation.
- **Section 6: Validation** – the OLLA target-BLER sweep and random/constant controls, ruling out mis-tuned-baseline and policy-collapse explanations.
- **Section 7: Discussion** – setpoint effects vs. learned adaptivity, implications for the RL-in-wireless literature, deployment cost-benefit (no calibration pass vs. no throughput gain).
- **Section 8: Conclusion.**

---

13 Key Code Snippets

13.1 HARQ Chase-Combining Transmission Physics

The shared per-TTI transmission function (`txStep_v7.m`), used identically by every trainer and by evaluation, showing how retransmissions are charged against throughput and how OLLA updates only on first-transmission feedback:

```
if env.harq_on && H.pend(i)
    % retransmission of the pending TB (no OLLA update)
    effS = Strue_row(i) + env.harq_comb_dB * H.nret(i);
    bl   = estimateBLER_v7(effS, H.mcs(i));
    if rand > bl
        tp(i) = H.tbs(i) * 1e-3;       % delivered at last
        H.pend(i) = false; H.nret(i) = 0;
    else
        H.nret(i) = H.nret(i) + 1;
        if H.nret(i) > env.harq_max_retx % give up, drop TB
            H.pend(i) = false; H.nret(i) = 0;
        end
    end
else
    % new data at the chosen CQI
    mcs = sent(i) - 1;
    bl  = estimateBLER_v7(Strue_row(i), mcs);
    TBS = tbsFromMCSandRB(mcs, rb(i));
    ack = rand > bl;
    if ack
        tp(i) = TBS * 1e-3;
        off(i) = off(i) + du;          % OLLA: first-Tx ACK
    else
        off(i) = off(i) - dd;          % OLLA: first-Tx NACK
        if env.harq_on
            H.pend(i) = true; H.nret(i) = 1;
            H.tbs(i) = TBS; H.mcs(i) = mcs;
        end
    end
end
```

13.2 Behavior-Cloning Warm Start (Tier-1 Fix)

The fine-tuning-preserving behavior-cloning loss used in `tier1b`, cross-entropy on FixedPU's per-class actions plus a soft entropy bonus to avoid saturating the clone:

```
function [aGrad, cGrad] = bc_grads(actorNet, criticNet, S, Atgt, R, entW)
    V = forward(criticNet, S);
    cLoss = mean((R - V).^2);

    logits = forward(actorNet, S);
    logp_all = logits - (max(logits,[],1) + log(sum(exp(logits - max(logits,[],1)),1)));
    mask = zeros(size(logits), 'like', extractdata(logits));
    for b = 1:size(S,2), mask(Atgt(b), b) = 1; end
    ce = -mean(sum(logp_all .* dlarray(mask), 1));

    entropy = -sum(exp(logp_all) .* logp_all, 1);
    aLoss = ce - entW*mean(entropy);   % entW = 0.03: keeps clone soft

    cGrad = dlgradient(cLoss, criticNet.Learnables, 'RetainData', true);
    aGrad = dlgradient(aLoss, actorNet.Learnables);
end
```